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 Studierichting: Informatica
 Oefeningenreeks 5D nr. 3.5

$$\sin(2x) = 2 \cot(x)$$

$$\Leftrightarrow \cancel{2} \sin(x) \cos(x) = \cancel{2} \frac{\cos(x)}{\sin(x)} \rightarrow \sin(x) \neq 0 \Rightarrow x \neq k\pi$$

$$\Leftrightarrow \sin^2(x) \cos(x) = \cos(x)$$

$$\Leftrightarrow -\cos(x) (-\sin^2(x) + 1) = 0$$

$$\Leftrightarrow -\cos^3(x) = 0$$

$$\Leftrightarrow \cos(x) = 0$$

$$\Leftrightarrow x = \frac{\pi}{2} + k\pi \text{ met } k \in \mathbb{Z}$$

$$\Leftrightarrow x = \frac{\pi}{2} \text{ of } \cancel{\frac{3\pi}{2}}, \text{ want } \left[\frac{\pi}{3}, \frac{2\pi}{3} \right]$$

$$\begin{aligned} S &= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (2 \cot(x) - \sin(2x)) dx + \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} (\sin(2x) - 2 \cot(x)) dx \\ &= 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot(x) dx - \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin(2x) dx + \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \sin(2x) dx - 2 \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \cot(x) dx \\ &= 2 [\ln |\sin(x)|]_{\frac{\pi}{3}}^{\frac{\pi}{2}} + \frac{1}{2} [\cos(2x)]_{\frac{\pi}{3}}^{\frac{\pi}{2}} - \frac{1}{2} [\cos(2x)]_{\frac{\pi}{2}}^{\frac{2\pi}{3}} - 2 [\ln |\sin(x)|]_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \\ &= 2 \left(\ln(1) - \ln\left(\frac{\sqrt{3}}{2}\right) \right) + \frac{1}{2} \left(-1 + \frac{1}{2} \right) - \frac{1}{2} \left(\frac{-1}{2} + 1 \right) - 2 \left(\ln\left(\frac{\sqrt{3}}{2}\right) - \ln(1) \right) \\ &= -2 \ln\left(\frac{\sqrt{3}}{2}\right) - \frac{1}{4} - \frac{1}{4} - 2 \ln\left(\frac{\sqrt{3}}{2}\right) \\ &= -4 \ln\left(\frac{\sqrt{3}}{2}\right) - \frac{1}{2} \\ &= -2 \ln\left(\frac{3}{4}\right) - \frac{1}{2}, \text{ want oppervlakte is positief} \end{aligned}$$

References

5D-3.5-1uko.de.vriendt₂.INF.